



University
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Workshop on Theory
in Computational Interaction

Active Inference as an HCI theory

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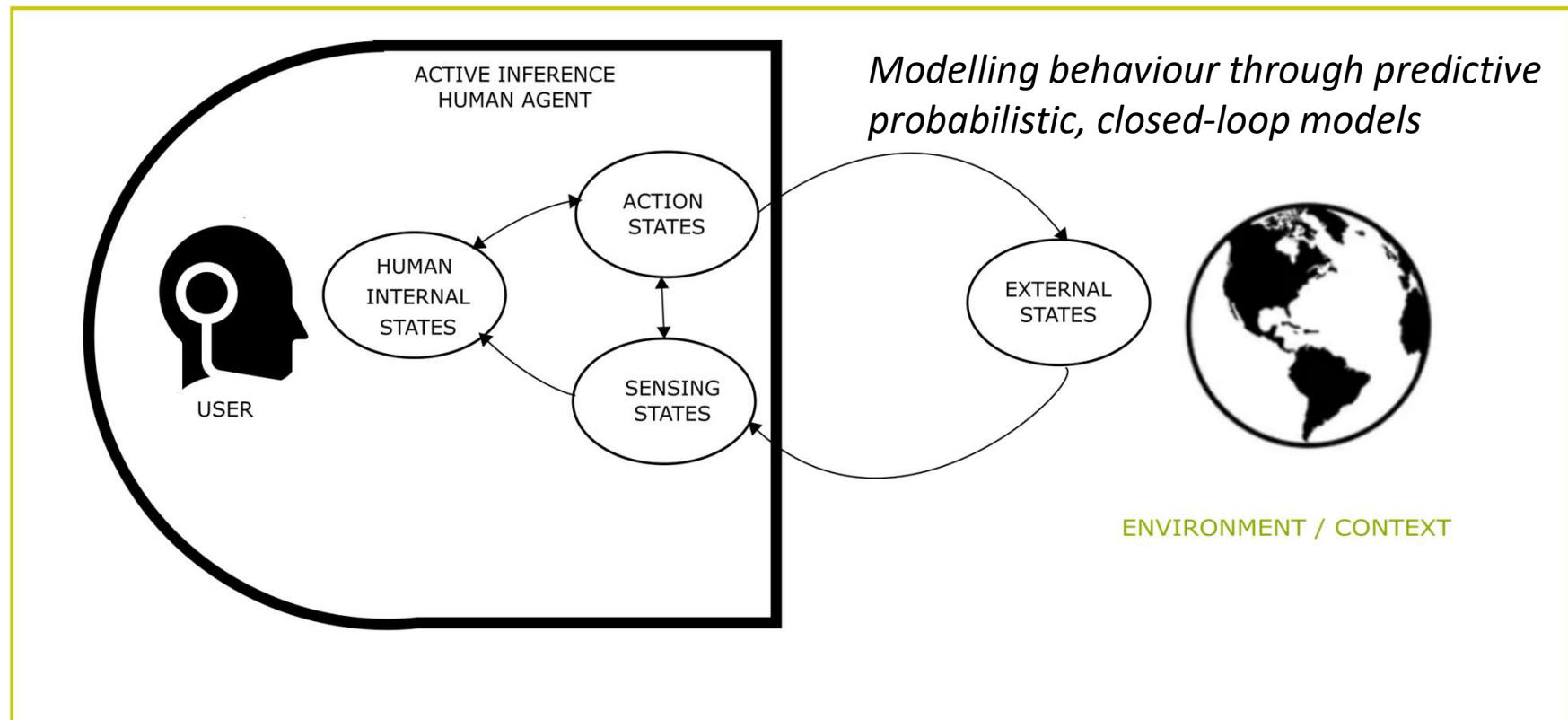
Active Inference and Human-Computer Interaction
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Active Inference is a closed-loop computational theoretical basis for understanding behaviour, based on agents with internal probabilistic generative models that encode their beliefs about how hidden states in their environment cause their sensations. We review *Active Inference* and how it could be applied to model the human-computer interaction loop. *Active Inference* provides a coherent framework for managing generative models of humans, their environments, sensors and interface components. It informs off-line design and supports real-time, online adaptation. It provides model-based explanations for behaviours observed in HCI, and new conceptual tools with the potential to measure important concepts such as agency and engagement in interaction. We discuss how *Active Inference* offers a new basis for a theory of interaction in HCI, tools for design of modern, complex sensor-based systems, and integration of artificial intelligence technologies, enabling it to cope with diversity in human users and contexts. We discuss the practical challenges in implementing such *Active Inference*-based systems.



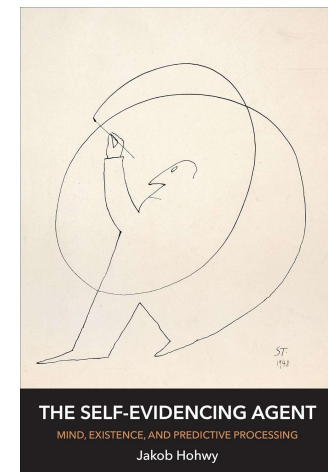
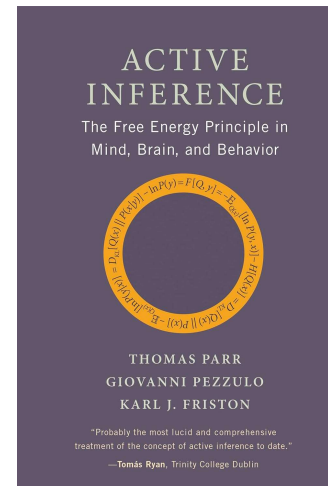
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1. How does Active Inference represent humans?

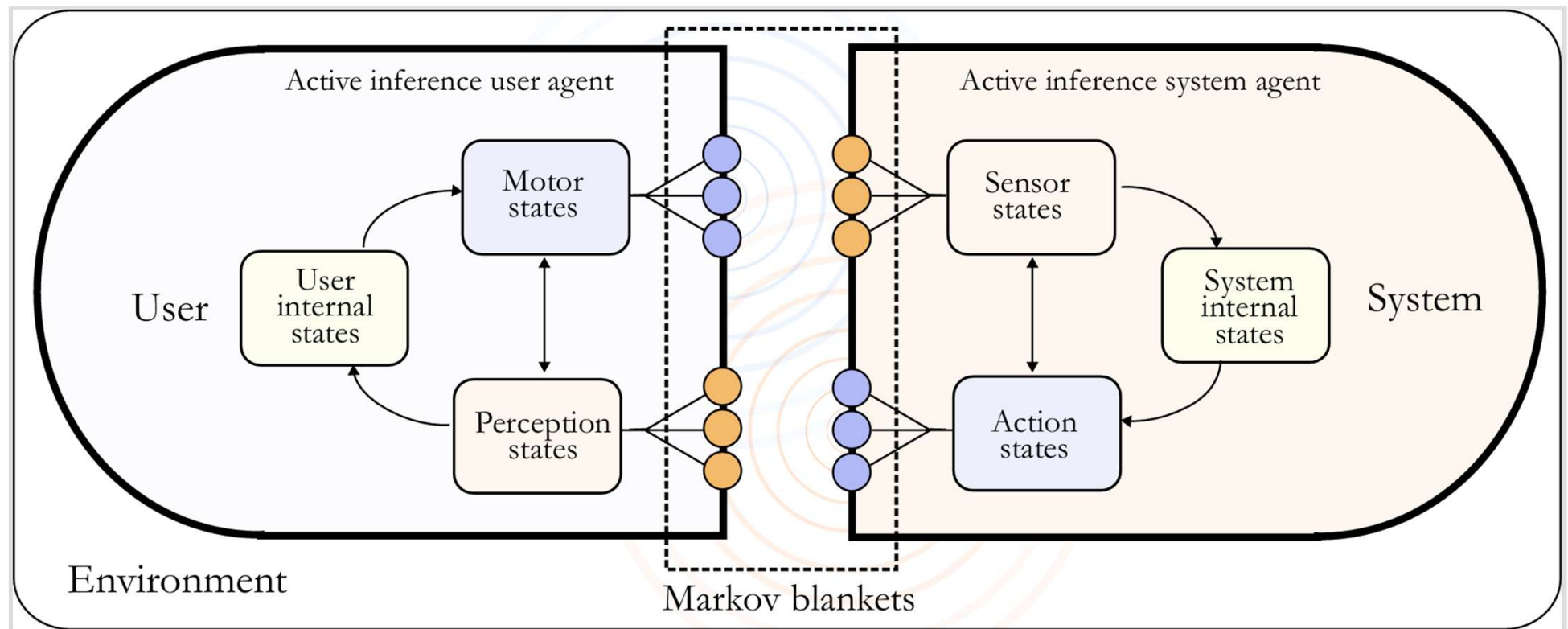


AIF - Fundamental theoretical motivations

- Friston argues that in a changeable world, active, self-organising systems exist by managing to occupy their expected states; i.e. they organise themselves and act in their environments to avoid surprising states, *given their model*.
 - I.e. Active Inference provides a Grand unifying theory for sentient agents
 - The closed-loop nature of interactions with an environment is central to the approach.
- Aligns with the *predictive processing* approach in neuroscience – we don't passively receive and process sensory input, we continually *generate* top-down models to *anticipate* sensory data. Prediction errors are used to infer the state of the world.
- Hohwy argues *Self-evidencing* occurs when organisms act in the world in ways that generate evidence for their existence, as in active inference.
 - AIF becomes a first-principles method for understanding the existence of agents...
- As with any good approach, it overlaps with many previous approaches:
 - Perceptual Control Theory, Model-based RL, Computational Rationality

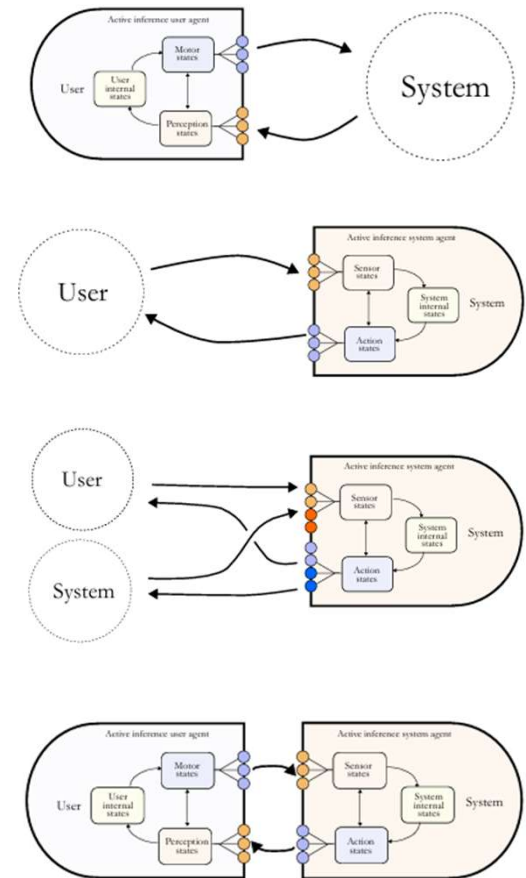
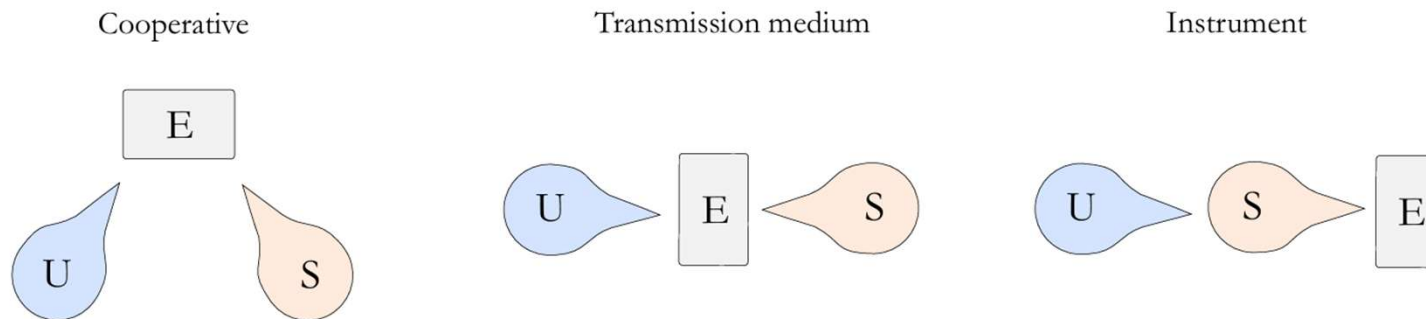


2 How does Active Inference represent the computer or AI?



3. What mechanisms link humans and computers?

- The closed-loop nature of an agent's interactions with an environment is central to the theoretical approach.
- Choice of mechanism will be critical in making the interaction manageable



OPTIMAL MECHANISM DESIGN

Latent sensing

Use closed-loop data-driven approaches to *learn* salient aspects of sensing inputs.

Continuously embedded state machines

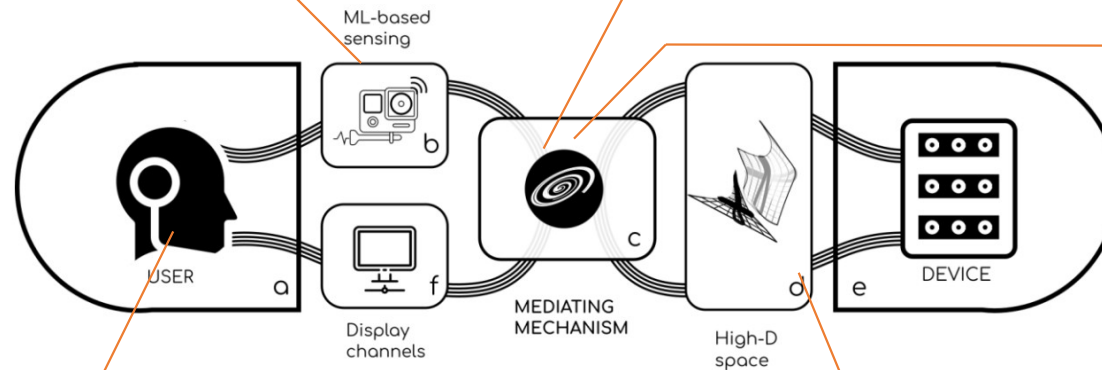
Bridge discrete-state and continuous control through phase-state-like systems.

Stable, malleable dynamics

Synthetic mechanics with designed feedback-control properties, parametrically configured.

Hierarchical intelligence

Slow-time modulation of fast-loop dynamics using higher-level context.



Simulated users

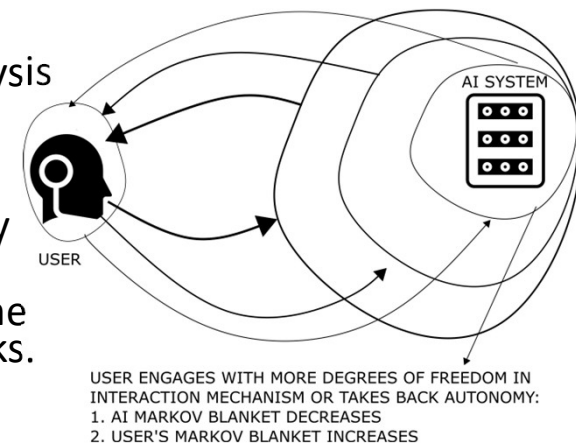
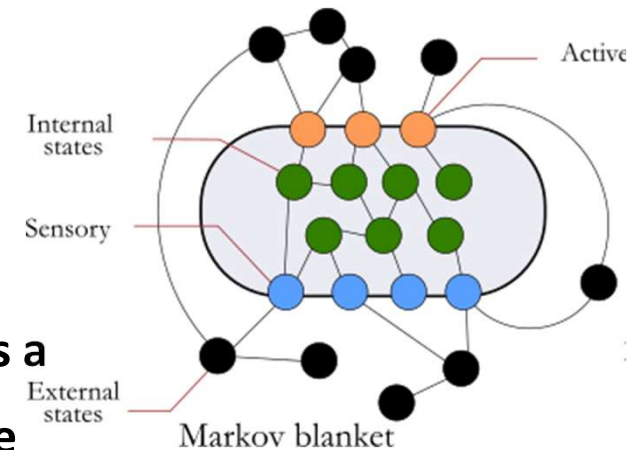
Apply biomechanical and perceptual simulation to afford *in silico* development and evaluation workflows.

A new interaction substrate

A *parametric* dynamical interaction which is **differentiable and optimisable**, bridging UI and human sensorimotor loops which affords intelligent continuous control beyond cursors and gestures.

Where are the boundaries in Human–Computer Interaction?

- Finding **meaningful boundaries between an agent and its environment is a core aspect of Active Inference** and is a **fundamental and controversial question** of general interest for HCI, especially when AI assistants can **take agency** from humans for certain tasks.
- The very name *Human–Computer Interaction* assumes a clear boundary between them!
 - A Markov blanket in the Active Inference sense between human and machine. (Clark's 'skin-bag')
 - Finding the boundaries between a system and its environment is essential for analysis of the system.
- ..but *Markov blankets change dynamically* as we give and take control
 - Technology can change the boundary. By giving a user a tool or assistive technology we transform their action and sensing capabilities.
 - Modern tools include context-sensitive algorithmic support from AI, augmenting the user cognitively, or relieving them of the need to attend to, or act on mundane tasks.
 - This can be seen as taking or augmenting **agency**



Slide 7

RM1

Integrate next slide bits

Roderick Murray-Smith, 2025-01-23T14:36:38.548

Theoretical questions for H-AI Interaction

- Humans and AI systems both act to maximise **predicted** benefit.
 - How do humans represent **the uncertainty about the expected response of the environment to an action with a tool**?
 - AI tools are **active, more complex, context-dependent, and hence less predictable** than passive tools.
 - AI context may not align with human-perceived context.
- How do humans and agents interact with the **Environment** and each other?
 - How do their **mutual models** affect their behaviour?
 - What are the **boundaries** between humans, tools and environment and how do these change?
 - Human and AI can act at a range of **hierarchical** levels.
- What are the **preferences/goals/reward structures** for humans and AI agents?
 - How do we define **Tasks** in a given Environment? How do these interact with preferences?
- Interaction
 - How is it theoretically defined? Can we measure it? Can we explain it?

4. Which computational problems in interaction can be rigorously defined using the theory?

- A computational (hence simulatable) approach to modelling intelligent behaviour and agency.
 - makes it a strong candidate for contributing to the study of human–computer interaction, where humans are intelligent agents, and we are increasingly attempting to build more intelligent computational systems to interact with.
- Explicit probabilistic models of uncertainty for environment, perception and action make it robust
- Its predictive nature allows it to explore and be proactive.
- It can be applied to low-level physical interactions or higher-level reasoning and decision tasks.
 - a basis for scientific, quantitative analysis & engineering of interactive systems, giving detailed predictions at a range of levels, with broad application.
 - an AIF approach to HCI emphasises explicit and precise description of the interaction in a form that can be analysed in silico

Challenges!

- Computation
 - AIF is computationally intensive, particularly in action rollouts, and engineering effort will be needed to balance planning with real-time requirements of online applications, for example via amortised algorithms
- Implementation
 - We don't have the software libraries and computational abstractions to build, reason about, and deploy interfaces.
 - We lack developed workflows to apply Active Inference to interaction
- Preference Modelling
 - novel approaches to preference are flexible but unfamiliar.
- Uncertainty & Prediction:
 - Representing uncertainty can make interfaces robust but users can struggle to comprehend uncertainty.
 - Prediction is vital for AIF and can mitigate latency in the interaction loop.
 - But prediction can be unsettling for users. The predictive, proactive nature may make AIF interfaces appear to be in the 'uncanny valley'.

Solutions..

- Amortisation
- Offline use can still be valuable.
- The agent's form can allow it to become a statistical model of its niche, embodying statistical regularities of its world.
- A pair of AIF agents will bring their joint system into a mutually predictable region of the state space, rather than having to accurately model a wide range of 'outlier' behaviour.
- Preferences
 - Will require developments in how preference priors should be elicited and validated,
 - how generalised preferences can be fused from individual preferences, and
 - possibly hierarchies of preferences
 - Augmenting sparse preference functions based on experience. (like value functions in RL)

Contrasts

Classical HCI

Emphasises tasks, input/output and metrics. Design is foregrounded as the way to imbue a system with desirable properties.

Maximises specific metrics.

Perceptual Control

Emphasises **homeostasis** of percepts in the now. Hold an observation at a set-point by rejecting disturbances.

Minimises error.

Reinforcement learning

Emphasises maximizing reward as a strategy to plan. Rewards are extrinsic and are associated to the environment.

Maximises reward.

Computational rationality

Emphasises bounded-optimal behaviour: the best a resource-limited agent can do at a task given cognitive/motor bounds. Scalar, extrinsic utility; user is the agent.

Maximises bounded-optimal utility.

Active Inference

Emphasises agents, not necessarily the user, minimizing expected free energy of uncertain future states. EFE combines exploitation and exploration.

Minimises surprise*.

* “Surprise” idiosyncratically defined to mean unexpected *and* undesirable in the long-term.